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SEPARATOR FOR WING TRAP

Abstract

The invention relates to a separator for a wing trap, which comprises a securing element and a support element. Both elements are articulated together at one end with a fold line that acts as a hinge between the two elements, allowing the securing element to fold over the support element. When folded, the upper face of the securing element aligns with the upper face of the support element, so that a wire can be inserted between the two faces. This wire is secured and allows for determining the opening distance of a wing trap.

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Background/Summary

BACKGROUND

1. Field of the Invention

[0001] This invention relates to the monitoring and control of pests, more specifically to an insect trap, and even more particularly to a separator used in a wing trap.

2. General Background of the Invention

[0002] Throughout time, humanity has attempted to control insects that are harmful to crops. However, the wide variety of insects and their resistance over the years forces us to improve our methods of controlling these insects. Consequently, there are more sophisticated control methods, including insect traps with very particular characteristics aimed at specific types of insects. In this context, insect traps are made in different types, allowing for increasingly precise control by having the most suitable trap for the insect to be controlled or monitored.

[0003] Monitoring and controlling insect populations can be carried out using insect traps, which come with various specifications depending on the type of insect to be controlled, as mentioned. Along these lines, improvements are made to the accessories present in each insect trap. Therefore, a trap separator has been developed, which improves the efficiency of insect traps. It is important to note that multiple varieties have been developed to cover this accessory but without achieving an insect trap separator like the one achieved in the present invention. It is important to look at different types of trap separators that have been implemented, which present different configurations compared to the one developed in our invention.

[0004] Here is shown the Patent Application US 2007/0094915 A1 by Plato et al., dated May 3, 2007, titled "Insect Trap for Capturing Numerous Species of the Order Lepidoptera and Method of Operation Thereof." In this trap, a separator for parts of the insect trap can be seen, which, through supports, performs the function of separating these parts. However, to perform this function, supports and a post where these supports will be placed are needed to achieve the separation of the trap parts. This invention has additional elements to the insect trap that make it difficult to arrange all the elements for its operation in elevated parts such as a tree.

[0005] Another example of such separators is found in the Utility Model KR 20100000227 U1, by Jeongsik Lee, dated Jan. 7, 2010, titled "Tunnel-Type Pest Trap." In this invention, a tunnel type is constructed with walls impregnated with an attractant. When assembling the tunnel, initially presented in a flat preform, holes are used to insert a cord that serves as support and controls the opening of the assembled tunnel, thus creating the required space for insects to enter the trap. However, it is not possible to reuse or open this tunnel-type trap once assembled, as doing so would lose the initial opening and contaminate the attractants due to the preform's arrangement.

[0006] Wire separators have also been made, which allow space for a wing trap. These separators are generally made manually in the field where these traps are to be placed. In this regard, these separators are practical in their execution, with the craftsmanship depending on the skill of the person responsible for this task. However, these separators do not standardize the openings of the traps, nor do they allow for reuse, as they generally become deformed. Currently, crudely bent wires are used. The bends in these wires or hooks are formed after being inserted into the upper part of the trap. This process is done manually, and the bends are inconsistent and difficult to form. This operation is very laborious and causes problems, not only in assembling the trap but also in failing to firmly secure the base of the trap. In situations with strong winds, these bases are lost due to not being firmly secured. Another way to form the wing trap is by using a pair of straws cut to a specific size. However, the disadvantage is that when the base of the trap, which is impregnated with insect-holding glue, needs to be discarded, it is necessary to completely disassemble the trap and separate both parts. The upper part (lid) is attached to the lower part (base) by a hook that passes through both parts.

[0007] From the background information mentioned and existing solutions, no wing trap separator with the characteristics we propose has been found. Therefore, we recognize the need for a wing

trap separator, leading to the development of the present invention, which is detailed below.
SUMMARY OF THE INVENTION
[0008] The present invention relates to a separator for a wing trap, which will improve the efficiency of wing traps, as this separator features characteristics that enhance the efficiency and quality of the performance of these traps.
[0009] This separator facilitates the assembly of the trap, in addition to controlling a specific separation required for the capture of moths and insect pests that are monitored and trapped in this type of trap.
[0010] This separation is important for the placement of attractants and the airflow that enters and disperses the active ingredients of these attractants.
[0011] Additionally, this specific separation allows insects to enter based on their size and flight behavior.
[0012] With our separator, the upper part (lid) is securely attached to the lower part (base) and can easily be detached from one another without having to disassemble the trap.
[0013] When between 50 to 100 traps are placed per hectare, it results in a time-saving advantage, which is an important economic factor at the end of the day.
[0014] Therefore, the object of the present invention is to provide a separator for wing traps that aids in the monitoring and control of insects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0015] To complement the description being made and to aid in a better understanding of the characteristics of the invention, the present description is accompanied by the following drawings, which are an integral part of it and have been illustrated and include, but are not limited to:
[0016] FIG. **1**. Shows a side view of the separator for the wing trap.
[0017] FIG. **2**. Shows a top view of the separator for the wing trap.
[0018] FIG. **3**. Shows a bottom view of the separator for the wing trap.
[0019] FIG. **4**. Shows a front view of the separator for the wing trap.
[0020] FIG. **5**. Shows an isometric top view of the separator for the wing trap.
[0021] FIG. **6**. Shows an isometric bottom view of the separator for the wing trap.
[0022] FIG. **7**. Shows an isometric front view in the closing position of the separator for the wing trap.
[0023] FIG. **8**. Shows a top view in the open position of the separator for the wing trap with the wire inserted.
[0024] FIG. **9**. Shows a side isometric view of the separator for the wing trap being folded with the wire inserted.
[0025] FIG. **10**. Shows a top view in the closed position of the separator for the wing trap with the wire inserted.
[0026] FIG. **11**. Shows a front view of the separator placed in the wing trap together with the wire.
DETAILED DESCRIPTION OF THE INVENTION
[0027] The following is a detailed description of the present invention with reference to the figures.
[0028] According to the present invention, the separator for the wing trap **1** comprises a securing element **20** and a support element **10** (see FIG. **1**). Both elements are articulated together at one contiguous end, thus forming a fold line **30** (see FIGS. **1** to **3**) that acts as a hinge between the securing element **20** and the support element **10**. This arrangement allows the securing element **20** to be folded over the support element **10**, or vice versa. When folded, the upper face **18** of the securing element aligns with the upper face **28** of the support element, as both faces have the same length and width. This alignment makes it possible to secure a wire **40** between the two surfaces.

The upper face **28** of the support element has a surface substantially equal to the upper face **18** of the securing element, allowing the piece to interact with a securing channel of the wire **11** against the wire support channel **21** (see FIG. **2**). These elements will secure the wire when folded.
[0029] The separator for the wing trap **1** includes two elements joined by a hinge or fold line **30**. The securing element **20** has the shape of a longitudinal box or a hollow rectangular prism, with an initial channel space **12** at one end near the fold line **30** (see FIGS. **2**, **3**, **5**, **6**, **8**, and **9**), and a rectangular projection at the opposite end of the securing element **20** forming a latch **14** (see FIGS. **1** and **5**). The arrangement of the initial channel space **12**, the wire securing channel **1**, and the wire exit guide **13** are aligned, allowing for the longitudinal insertion of a wire **40** to open and close the securing element **20** against the support element **10**, as seen in FIGS. **8**, **10**, and **1**. It is worth mentioning that these three parts—the initial channel space **12**, the wire securing channel **1**, and the wire exit guide **13**—are aligned below the upper face **18** of the securing element **20**, enabling the placement of the wire **40** in this alignment without protruding from the upper face **18**.
[0030] The separator for the wing trap **1**, with its support element **10**, has the shape of a longitudinal rectangular plate. On its upper face **28**, it features a wire support channel **21** (see FIG. **2**), which does not span the entire length of the surface **28**, leaving a space **22** between the wire support channel **21** and the fold line **30**. Similarly, at the opposite end of the wire support channel **21**, there is a free space that does not reach the end of the support element **10**. These separations allow for the proper manipulation and bending when inserting the wire into the wire support channel **21**. Once the wire **40** is mounted on the support channel **21**, as seen in FIGS. **8** and **9**, the support element **10** can be folded with the securing element **20** in such a way that the wire **40** and the support channel **21** are enclosed within the cavity of the securing element **20**.
[0031] In FIGS. **5** and **6**, it is shown that at the opposite end of the fold line **30** of the support element **10**, there is a tab support **25** inclined towards the initial channel space **12** and the final channel space **23**. This tab support **25** has a locking slot **24**, which allows the insertion of the latch **14**. Additionally, on the tab support **25**, there is a pin tab **27** located at the end, with a slight inclination similar to the tab support **25**. The pin tab **27** has an internal pin **26**, which allows it to engage with the latch **17** (see FIGS. **3**, **5**, and **7**). When folding the securing channel **1** against the support element **10**, specifically against the wire support channel **21**, as shown in FIG. **7**, the engagement part of the latch **17** locks with the pin **26** when both parts are folded, thus joining both parts as shown in FIGS. **4**, **10**, and **11**. It is worth noting that the pin can be manipulated by a user by applying a small force “F**1**” in the direction of the pin tab **27** when closing and a force “F” when opening the pin, as seen in FIG. **6**.
[0032] In FIGS. **6** and **7**, the numeral **15** indicates the separation tab. This tab is located on the lower face **19** of the securing element **20**, situated at an end near its fold line **30**. The separation tab **15** is arranged perpendicularly to the lower face **19** of the securing element **20**. This separation tab allows for a specific separation between the wings of the wing trap to create a fixed opening for insect entry. The separation tab can be adjusted along the lower face **19** of the securing element **20** to achieve openings according to the needs of the insect to be captured.
[0033] The purpose of the separation tab **15** is to stabilize and eliminate the movement of the trap base, keeping it in a constant position.
[0034] The space between the channel fold **22** and the fold line **30** has a specific measurement to accommodate the edge of the trap base, which is inserted between the support element **10** and the securing element **20** of the separator.
[0035] FIGS. **8** to **10** show the arrangement of the wire **40** passing through the locking slot **24**, then being placed on the support channel **21** of the support element **10**, and finally exiting through the initial channel space **12** of the securing element **20**. Once positioned, the wing trap separator is folded and secured with the pin in the latch engagement, firmly holding the wire that has passed through the wing trap separator **1**. This separator will allow for a predetermined distance achieved between the separation tab **15** and the tab support **25**, which will establish the separation of the

wings of a wing trap.
[0036] FIG. **1** shows how the separator **1** adapts to the upper part of the wing trap, held by wires at both its upper and lower parts.
[0037] To better provide an understanding of the invention, a list of parts that constitute the separator for the wing trap is presented. [0038] **1** Separator for Wing Trap [0039] **20** Securing Element [0040] **11** Wire Securing Channel [0041] **12** Initial Channel Space [0042] **13** Final Channel Space [0043] **14** Latch [0044] **15** Separation Tab [0045] **16** Hollow Channel Space [0046] **17** Latch Engagemen [0047] **18** Upper Face of Securing Element [0048] **19** Lower Face of Securing Element [0049] **10** Support Element [0050] **21** Wire Support Channel [0051] **22** Space Between Channel and Fold Line [0052] **23** Final Channel Space [0053] **24** Locking Slot [0054] **25** Tab Support [0055] **26** Pin [0056] **27** Pin Tab [0057] **28** Upper Face of Support Element [0058] **29** Lower Face of Support Element [0059] **30** Fold Line [0060] **40** Wire [0061] **50** Upper Lid of Wing Trap [0062] F**1** Closing Force of Separator [0063] F Opening Force of Separator

Claims

1. A separator for a wing trap, comprising: A securing element (**20**) in the shape of a longitudinal box, with an initial channel space (**12**) at one end near a fold line (**30**) and a rectangular projection at the opposite end of said securing element forming a latch (**14**), where the arrangement of the initial channel space (**12**), a wire exit guide (**13**) arranged in the latch (**14**) on the upper part of the securing element (**20**), and a securing channel (**11**) are aligned; A support element (**10**) in the shape of a longitudinal rectangular plate having on its upper surface (**28**) a wire support channel (**21**), which does not span the entire length of the surface as it has spaces at the start of the channel (**22**) and the end of the channel (**23**), as well as a pin tab (**27**). The securing element (**20**) and the support element (**10**) are aligned at the upper face of the securing element (**18**) and the upper face of the support element (**28**) when folding the wing trap separator (**1**) through the fold line (**30**), and securing the wing trap separator by means of the latch (**14**) and the pin tab (**27**).
2. A separator for a wing trap according to claim 1, characterized in that the support element (**10**) has, at the opposite end of the fold line (**30**) of the support element (**10**), a tab support (**25**) arranged inclined from the initial channel space (**22**) towards the final channel space (**23**) with respect to the upper face of the support element (**28**). This tab support (**25**) has a locking slot (**24**).
3. A separator for a wing trap according to claim 1, characterized in that the tab support (**25**) has a pin tab (**27**) located at the end with a slight inclination in the same direction as the tab support (**25**). The pin tab (**27**) has an internal pin (**26**), which allows it to engage a latch engagement or pin (**17**). Thus, when folding the securing element (**20**) against the support element (**10**), the latch engagement part (**17**) locks with the pin (**26**), thereby joining both parts.
4. A separator for a wing trap according to claim 1, characterized in that the pin (**26**) can be manipulated by a user by applying a small force "F**1**" in the direction of the pin tab (**27**) when it is to be closed, and a force "F" when the pin is to be opened.
5. A separator for a wing trap according to claim 1, characterized in that the separation tab (**15**) is located on the lower face (**19**) of the securing element (**20**) near its fold line (**30**), where the separation tab (**15**) is arranged perpendicularly to the lower face (**19**) of the securing element (**20**).
6. A separator for a wing trap according to claim 1, characterized in that the wire (**40**) passes through the locking slot (**24**), then is placed on the support channel (**21**) of the support element (**10**), and finally exits through the initial channel space (**12**) of the securing element (**20**). Once positioned, the wing trap separator is folded and secured by means of the pin in the latch engagement, firmly holding the wire that has passed through the wing trap separator (**1**).
7. A separator for a wing trap according to claim 1, characterized in that the separator performs the separation through the separation tab (**15**) and the tab support (**25**), which allow for the separation of the wings of a wing trap.

8. A separator for a wing trap according to claim 1, characterized in that the separation tab (**15**) can vary its position along the lower face (**19**) of the securing element (**20**) to achieve openings according to the needs of the insect to be captured.